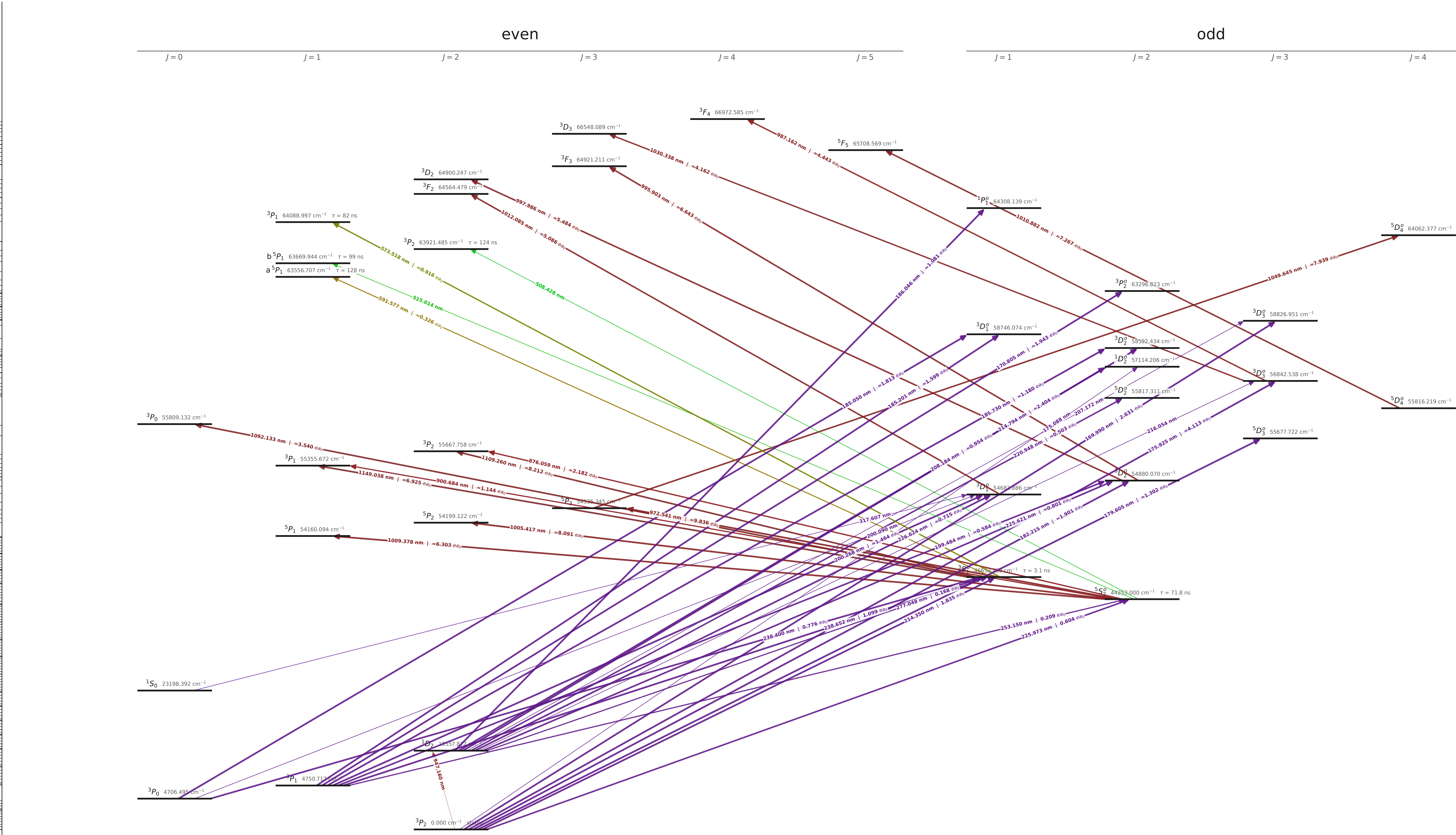


Te (Te I). Strongest classified transitions below 2 μm (NIST): 36 levels, 47 lines.

Te⁺ ionisation limit 72669.01 cm⁻¹ = 9.00981 eV (137.610 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle V | \mathbf{e} \cdot \mathbf{r} | J' \rangle$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U er U\rangle$ (ea)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$^3P_2 - ^3S_1^o$	214.3497	214.2822	1398.6139	1.835	2.310e+08	51.34	71.6	first step of the ISOLDE RILIS resonance-ionization schem...
$^3S_1^o - ^3P_1$	573.5175	573.3585	522.7259	≈ 0.916	3.006e+06	1.941	24.6	second step of the ISOLDE RILIS resonance-ionization sche...
$^3S_1^o - a\ ^3P_1$	591.5770	591.4132	506.7682	≈ 0.326	3.470e+05	1.243	4.44	second resonant step

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3P_2	-998(14)		- F=3/2: +1497.00 F=5/2: -998.00
$^5S_2^o$	1010.4(13)		- F=3/2: -1515.60 F=5/2: +1010.40
$^3S_1^o$	943(20)		- F=1/2: -943.00 F=3/2: +471.50

Measured A, B constants; sources in the data file.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^3P_2 - ^3S_1^o$	214.350	120-130	-1029(11)
$^3P_2 - ^3S_2^o$	214.350	122-130	-829(13)
$^3P_2 - ^3S_1^o$	214.350	124-130	-625(13)
$^3P_2 - ^3S_2^o$	214.350	126-130	-426(11)
$^3P_2 - ^3S_1^o$	214.350	128-130	-211(11)
$^3P_2 - ^3S_2^o$	214.350	132-130	214(15)
$^3P_2 - ^3S_1^o$	214.350	134-130	402(15)
$^3P_2 - ^3S_2^o$	214.350	136-130	1242(21)
$^3P_2 - ^3S_2^o$	214.350	123-130	-815(15)
$^3P_2 - ^3S_1^o$	214.350	125-130	-594(19)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$